

Conservation and Restoration of a Rare Historical Crochet Tablecloth Embroidered with Metal Threads

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ترميم و صيانة مفرش طاولة تاريخي نادر مصنع بتقنية الكروشيه ومطرز بالخیوط المعدنية

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Abstract:

Textiles represent one of the most important elements in the study of art history and ancient civilizations, as they embody the social, economic, and cultural development of different periods. These artifacts act as tangible evidence of the weaving and embroidery techniques, the materials employed, and the craftsmanship that characterized various historical eras. This research presents the examination, analysis, and conservation of a historical crochet table cover embroidered with metallic threads, housed at the Beit El-Umma Museum. The artifact's condition was thoroughly documented through detailed visual observation, high-resolution photography, and a comprehensive condition and damage mapping process using AutoCAD. Analytical techniques, including Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX), were employed to accurately identify the fiber and metal compositions and to investigate the deterioration mechanisms affecting the object.

The conservation approach involved both mechanical and chemical cleaning, reinforcement of structurally weakened areas with silk crepe-line, and mounting on a custom-designed support using specialized conservation stitching methods. This study proposes a reproducible conservation methodology for composite textiles combining organic fibers and metallic elements, adhering to international conservation ethics of minimal intervention, reversibility, and preventive care. The results contribute to the long-term preservation, study, and presentation of complex historic textiles within museum collections.

Keywords: Consolidation - Sirma - Textile - Scanning Electron Microscopy (SEM) - Conservation

الملخص:

تمثل المنسوجات أحد أهم العناصر في دراسة تاريخ الفن والحضارات القديمة، إذ تجسد التطور الاجتماعي والاقتصادي والثقافي لمختلف العصور. كما تُعد هذه القطع الأثرية دليلاً ملموساً على تقنيات النسيج والتطريز، والمواد المستخدمة، ومهارة الحرفيين التي ميّزت الفترات التاريخية المتعاقبة.

يقدم هذا البحث دراسة تشمل عمليات الفحص والتحليل والترميم لمفرش طاولة تاريخي من الكروشيه مطرّز بخيوط معدنية، ومحفوظ بمتحف بيت الأمة. تم توثيق حالة القطعة بدقة من خلال الملاحظة البصرية التفصيلية، والتصوير الفوتوغرافي عالي الدقة، وإعداد خريطة شاملة للحالة ومظاهر التلف باستخدام برنامج AutoCAD.

وقد استُخدمت تقنيات تحليلية متقدمة مثل الميكروسكوب الإلكتروني الماسح (SEM) والتحليل الطيفي للأشعة السينية المشتتة للطاقة (EDX)، لتحديد تركيب الألياف والمعادن بدقة، ودراسة أسباب التدهور التي أثّرت على القطعة. شملت خطوات الصيانة التنظيف الميكانيكي والكيميائي، وتدعيم الأجزاء الضعيفة هيكلياً باستخدام نسيج الحرير الكريبيلين، وتثبيت المفرش على قاعدة مصممة خصيصاً له باستخدام طرق خياطة متخصصة. يقترح هذا البحث إحدى طرق الحفظ القابلة للتطبيق على المنسوجات المركبة التي تجمع بين الألياف العضوية والعناصر المعدنية، وفقاً لقواعد الحفظ الدولية التي تقوم على مبدأ أقل تدخل ممكن، والصيانة الوقائية. تهدف الدراسة الحفاظ على المنسوجات التاريخية المركبة ودراستها وعرضها على المدى الطويل ضمن مجموعات المتاحف.

الكلمات الدالة: التقوية، السيرما، المنسوجات، الميكروسكوب الإلكتروني الماسح، الصيانة.

1. Introduction

The conservation of historical textiles requires exceptional precision and a highly systematic approach, as these artifacts represent a physical manifestation of human civilization, carrying not only artistic and aesthetic value but also functioning as critical cultural artifacts that provide essential data through detailed material and technical analysis¹.

Preserving and restoring these objects safeguards the historical and visual memory of societies, while offering invaluable insights into traditional craftsmanship, the raw materials employed, and the cultural and social contexts in which they were produced². Conservation of historical composite objects such as textiles decorated with metal threads is a complex process³. This is because materials and methods that are suitable to conserve metal threads may be harmful for textiles⁴. Therefore, it is essential to develop conservation processes and materials that respect the integrity of the composite object and can preserve all components⁵.

These composite objects deteriorate due to the exposure to unstable environmental conditions such as inappropriate temperature, humidity, and light^{6,7,8}. Also, pollutants, microorganisms and insects play a role in deterioration of these objects⁹. Deterioration of textiles materials causes breakdown of the molecular structure, which turn results in the loss of strength and general durability, in discolorations, fading and affects the

¹ O. Abdel-Kareem, *The Guide in Treatment and Conservation of Ancient Textiles*, Cairo, 2002.

² O. Abdel-Kareem, "Textile Conservation Past, present and future," *Advanced Research in Conservation Science*, vol. 2, no. 2, pp. 1-15, 2021.

³ O. Abdel-Kareem, "Investigation and Conservation of a Historical Woman's Coat Decorated with Fur Parts," *Journal of textile and apparel technology and management*, vol. 7, no. (2), pp. 1-12, 2011.

⁴ O. Abdel-Kareem, "Challenges in the Conservation of Composite Artefacts, Part I: Establishing a Strategic Plan for Conservation of a Historical Woman's Coat," in *Proceedings 4th International Congress on "Science and Technology for Safeguard of Culture Heritage in the Mediterranean Basin"*, Cairo, 6th -8th December 2009A.

⁵ O. Abdel-Kareem, *Principles of Conservation and Restoration of Archaeological Organic Materials, textiles, wood, leather, manuscripts, boons, basketry and feathers*, Cairo: Cairo University Center for the Printing and the Publishing, 2015, p. 428.

⁶ Annu, A. S. Yadav, N. M. Rose and S. Rukhaya, "Importance of Textile Conservation in Cultural Heritage Management: A Review," *International Journal of Research Publication and Reviews*, vol. 6, no. 5, pp. 8938-8943, 2025

⁷ H. A. Al-Gaoudi and N. M. Aly, "The Characterization Of Some Ancient Egyptian Funerary Linens From The Twenty-First Dynasty Discovered In The Bab El-Gasus Excavation," *Journal Of Egyptian Archaeology*, vol. 107, no. 1-2, p. 115–128, 2021.

⁸ O. Abdel-Kareem, *Principles of Conservation and Restoration of Archaeological Organic Materials, textiles, wood, leather, manuscripts, boons, basketry and feathers*, Cairo: Cairo University Center for the Printing and the Publishing, 2015, p. 428.

⁹ Annu, A. S. Yadav, N. M. Rose and S. Rukhaya, "Importance of Textile Conservation in Cultural Heritage Management: A Review," *International Journal of Research Publication and Reviews*, vol. 6, no. 5, pp. 8938-8943, 2025.

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appearance of the material. The degradation of the metallic threads, common degradation products include copper oxides, basic copper sulphates, copper chloride and basic copper carbonates. Chlorine-and sulphur-based compounds are frequently derived from degradation compounds, and they show a lower rate of decrease than other contamination residues¹⁰.

The conservation of a historical textile object decorated with metal threads requires a comprehensive and structured methodology that begins with a detailed diagnostic assessment of its condition ¹¹, followed by the precise identification of its original components, including fibers, dyes, and weave structures—as well as an in-depth analysis of deterioration factors. Based on these findings, appropriate cleaning methods, stabilization treatments, and structural support systems are carefully selected to maintain authenticity while ensuring reversibility and enabling long-term monitoring¹². Cleaning of textile artifacts enhances the appearance of the object and aims at prolonging the life of an object¹³.

This study presents the examination, analysis, and full conservation process carried out on an embroidered table cover from the collection of the Beit El-Umma Museum, part of the Fine Arts Sector of the Ministry of Culture, registered under inventory number (273). Prior to treatment, the textile underwent a detailed visual examination, documentation through high-resolution photography, and the creation of a damage map using AutoCAD. Advanced instrumental techniques, including Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX), were employed to identify its fiber composition, metallic threads, and corrosion products, which informed the subsequent cleaning, stabilization, and restoration procedures.

2. Material and methods:

2.1 Material

the artifact examined in this study dates to the first half of the 20th century (1902–1944). It is a rectangular crochet tablecloth decorated with meticulously crafted metallic thread embroidery on an undyed linen ground fabric. The piece is framed by an elaborate crochet border made of undyed cotton fibers, showcasing a rich variety of botanical and geometric motifs.

¹⁰ A. Duran, R. Perez-Maquedab and J. L. Perez-Rodriguez, "Degradation processes of historic metal threads used in some Spanish and Portuguese ornamentation pieces," *Journal of Cultural Heritage*, vol. 36, pp. 135-142, 2019.

¹¹ O. Abdel Kareem and R. Alfaisal, "Treatment, Conservation and Restoration of the Bedouin died Textiles in the museum of Jordanian Heritage," *Mediterranean Archaeology and Archaeometry*, vol. 10, no. 1, p. 25-36, 2010.

¹² H. E. Ahmed, "Preservation and Conservation Strategies of Historical Textiles in Excavation: A Case Study of Egyptian Excavation," *Trends in Textile Engineering & Fashion Technology*, vol. 3, no. 4, 2018.

¹³ D. Singhal and S. Bhagat, "Cleaning Of Metal Threads In Brocade Artifacts," *Research journal of Social Science and Management*, vol. 07, no. 011, pp. 154-160, 2018.

The textile is officially registered under inventory number (273), with overall dimensions of 65 × 150 cm and inner textile dimensions of 45 × 110 cm. This table cover is an exceptional example of textile artistry, characterized by its refined embroidery, sophisticated crochet craftsmanship, and stylistic diversity. It reflects high levels of technical skill and aesthetic sophistication in both design and execution, making it a unique representation of artistic and decorative traditions from its period (Fig. 1).

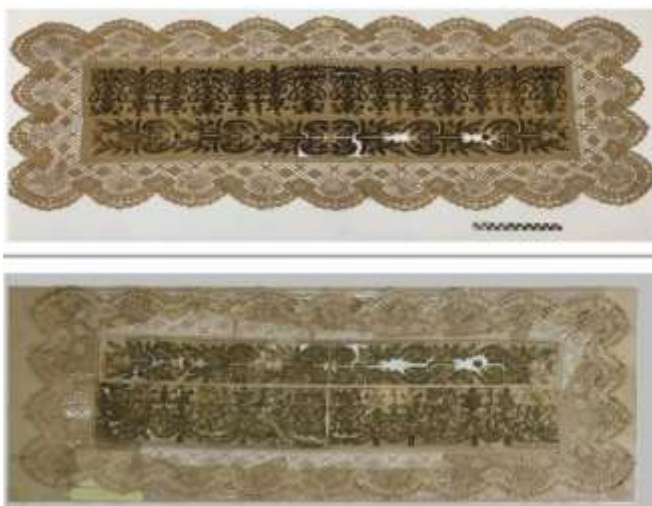


Figure (1): textile cover with a metallic thread embroidery, showcasing both its front and back.

The materials used in the conservation process included a chemical cleaning solution formulated for the treatment of metallic embroidery threads. The solution consisted of Rochelle salt solution (5 g sodium hydroxide and 15 g sodium potassium tartrate), sodium sesquicarbonate solution (2 g sodium carbonate and 2 g sodium bicarbonate), 1% EDTA, 1% Nonylphenol Ethoxylate (neutral industrial detergent), and 10% isopropyl alcohol, all diluted in distilled water. For surface consolidation and protection of the metallic components, Paraloid B-72 at 4% concentration was combined with 1% benzotriazole. Silk crepeline was applied to reinforce structurally weakened areas of the textile. The mounting system consisted of a treated acid-free pinewood frame and a plain-weave raw linen fabric (1:1) used as the primary support.

2.2 methods

2.2. 1. Documentation

The process of examination, recording, and documentation represents a fundamental preliminary stage in textile conservation. It provides a clear reference point for assessing the condition of the artifact prior to treatment and serves as a benchmark for evaluating the effectiveness and potential changes resulting from conservation interventions. Furthermore, it plays a crucial role in guiding decision-making, helping conservators select the most appropriate treatment strategies, techniques, and materials for restoration^{14, 15} In this study, the tablecloth was documented through three complementary methods:

Photographic documentation using a professional camera to capture high-resolution images of the textile's overall appearance and condition. Manual recording through a

¹⁴ S. Landi, *The Textile Conservator's Manual*, Butterworth & Heinemann., 1998.

¹⁵ J. Robinson and T. Pardoe, *An illustrated guide to the care of costume and textile collections*. London: Museums & Galleries Commission, museums & Galleries Commission., 2000.

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1:1 scale drawing, prepared by tracing the object's outline and details on a polyethylene sheet after carefully securing the piece on a Medium-Density Fiberboard (MDF) support .Digital technical documentation using AutoCAD software to create a precise deterioration map, visually representing areas of deterioration and guiding treatment planning.

2.2.2. Digital Microscope Examination

Microscopic analysis was conducted using a Leica M300 floor-mounted digital microscope equipped with an integrated digital camera and 150-watt fiber-optic coaxial lighting. The analysis was performed in the Conservation Laboratory of the Mahmoud Khalil Museum, affiliated with the Fine Arts Sector.

2.2.3. Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX)

Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX) is one of the most common techniques used in investigation of metal threads¹⁶. A Scanning Electron Microscope (SEM) (Model: FEI Quanta 3D 200i) equipped with an EDX unit (Thermo Fisher Pathfinder, Energy-Dispersive X-ray Analysis), housed at the Grand Egyptian Museum Conservation Center, was used to investigate the morphological structure and elemental composition of selected samples. The analyses were conducted under an accelerating voltage of 20–30 kV, with a 1 nm resolution gun, and at various magnifications utilizing both Secondary Electron Detector (SED) and Back-Scattered Electron Detector (BSED) imaging modes.

A total of three samples were examined: two taken from the linen ground fabric and one from the metallic sirma threads. Samples were mounted on standard-diameter aluminum stubs using carbon double-sided conductive tape, following standard SEM sample preparation protocols to ensure imaging accuracy and clarity.

3. Results and discussions

3.1 Visual examination and stylistic analysis:

The decorative design of the table cover is characterized by the use of simple, repetitive ornamental patterns. In the central section, created with the sirma embroidery technique, a composition of alternating composite patterns is employed, while the outer border, executed in crochet, features grid-like ornamentation combined with regular and alternating repetition of simpler patterns.

¹⁶ O. Abdel-Kareem, "Estimating the Condition of the Metal Embroidery Threads in Ottoman Textiles in Al-Manyal Palace Museum in Cairo," *Adumatu*, no. No. 19, pp. pp. 7-16, 2009B.

This artifact integrates two distinct fiber types and decorative techniques. The outer border is made from undyed cotton fibers, spun into yarn and worked entirely by hand using needlework crochet, while the inner section is woven from undyed linen fibers and richly decorated with metallic threads (sirma technique), forming a variety of botanical patterns (Fig. 2).

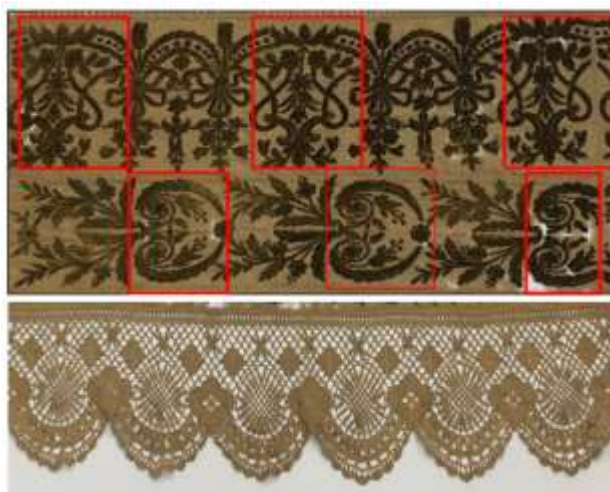


Figure (2): illustrates the artistic technique employed (the repetition of alternating composite ornamental motifs) and the (sirma) embroidery."

The metallic embroidery threads exhibit areas of weakness and corrosion layers, while the outer crochet border remains in relatively good condition. All forms of damage were thoroughly documented using AutoCAD software, resulting in a detailed damage map of the table cover (Fig. 4).



Figure (3): shows the various forms of deterioration on the piece, including severe tears and corrosion of the sirma threads.

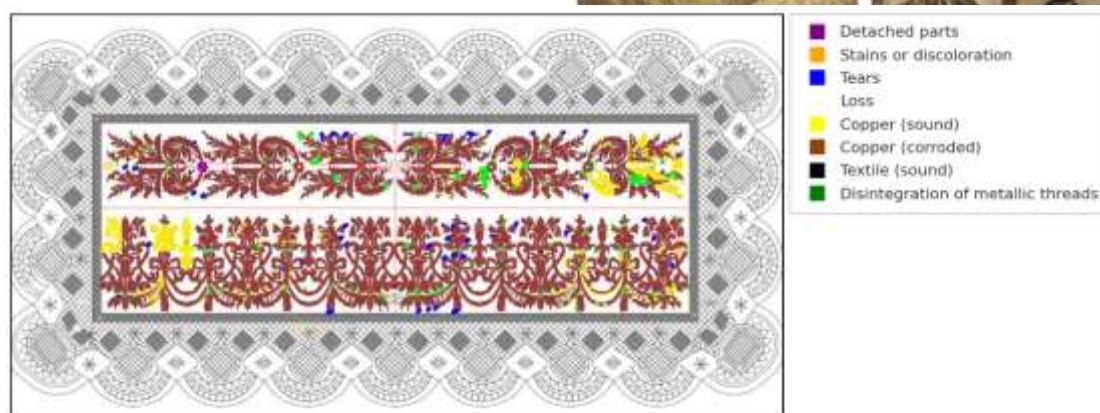


Fig. (4). Deterioration, illustrating all previously described deterioration features, accompanied by a color-coded key

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3.2 Digital microscope

Technical examination using the digital microscope at magnifications ranging from $\times 40$ to $\times 150$ revealed several key details, including the weave structure of the textile, which was identified as a 2/1 twill weave. The thread density was determined to be 24 warp threads and 18 weft threads per square centimeter (**Fig. 5**).



Fig. (5) Weave structure of the table cover, executed in a 2/1 twill pattern

Microscopic analysis of the metallic sirma threads further demonstrated the extent of corrosion accumulation and clustering on the metal threads, as well as their severe degradation and fiber disintegration, as illustrated in (**Fig. 6**).



Fig. (6) Accumulation of corrosion on sirma threads used in the decoration.

3.3 SEM and EDX

The first sample's fibers were examined using a Scanning Electron Microscope (SEM), and analysis of the longitudinal section confirmed that the warp threads are composed of (linen) fibers¹⁷. This is clearly visible in (**Fig. 7a**) at a magnification of $\times 2000$, where the distinctive structure and morphological features of fibers can be observed. Examination of the longitudinal section of the second sample revealed that the weft threads are made of cotton fibers, as indicated by their characteristic twisted, ribbon like morphology, which is clearly visible in (**Fig. 7b**) at $\times 2000$ magnification.

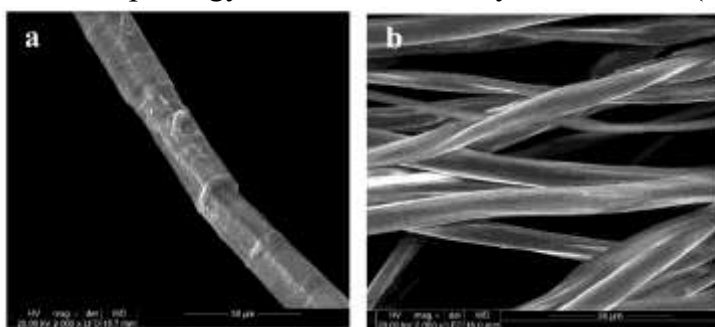


Fig. (7) Scanning Electron Microscope (SEM) images identifying the fiber types used in the table cover: (a) linen fibers, (b) cotton fiber

The third sample, taken from the metallic sirma threads, was analyzed using EDX (Energy-Dispersive X-ray Spectroscopy). The results revealed that copper (Cu) is the

¹⁷ H. A. Al-Gaoudi and N. M. Aly, "The Characterization Of Some Ancient Egyptian Funerary Linens From The Twenty-First Dynasty Discovered In The Bab El-Gasus Excavation," *Journal Of Egyptian Archaeology*, vol. 107, no. 1-2, p. 115–128, 2021.

primary constituent of the metal threads, accounting for 65.4 wt.%, indicating that the threads were manufactured primarily from copper or historically common copper alloys traditionally used in decorative textiles and jewelry. The oxygen content was measured at 13.3 wt.%, confirming the presence of copper oxides (Cu_2O and CuO), formed naturally due to prolonged exposure to moisture and atmospheric oxygen. The relatively high oxygen percentage suggests that these oxide layers are stable, forming part of the corrosion patina (**Fig. 8**)¹⁸.

Additionally, traces of chloride (Cl) and bromide (Br) were detected in low concentrations, suggesting that the textile and its sirma threads had been exposed to a saline or humid environment conducive to the development of bronze disease. This active form of corrosion results from the interaction of chloride ions with copper, leading to the formation of corrosion products such as atacamite and paratacamite¹⁹.

The detailed examination of the artifact, supported by visual inspection and instrumental analysis, demonstrated that the museum environment was a significant factor in its deterioration. Inappropriate display conditions, fluctuations in temperature, and exposure to dust, pollutants, and environmental factors have contributed to the oxidation of metallic threads and the development of corrosion layers. The textile exhibits structural fragility, weakness, and areas of material loss, while the fibers show signs of disintegration and lack of cohesion, confirming the urgent need for conservation and stabilization treatments.

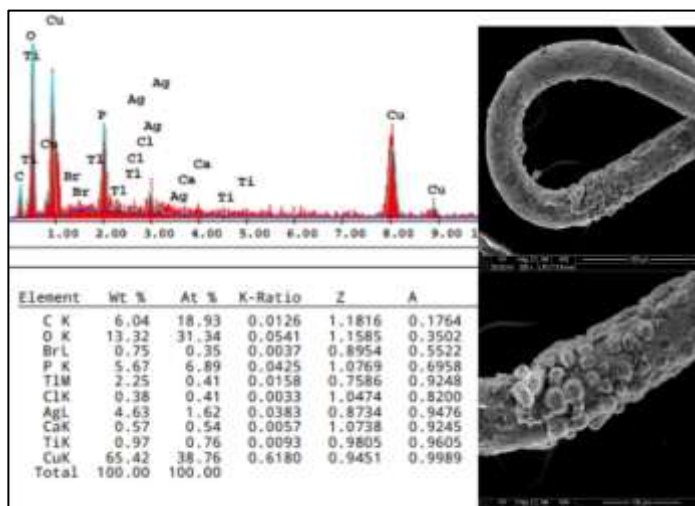


Fig. (8) SEM image of the sirma threads and EDX analysis showing the elemental composition of the sample

4. Conservation Process

4.1 Cleaning processes

Cleaning of historic textiles is an essential step which not only helps to prolong the life of the textile, but also removes the decaying material to some extent²⁰ There are many

¹⁸ R. Jiao, F. Sun and J. Li, "Application of low-temperature plasma for the removal of copper chloride layers on bronze Wares," *Heritage Science*, vol. 11, 2023.

¹⁹ H. Li, Z. Zhang, H. Guo, C. Ren, C. Liu and L. Xiang, "Recent Progress on Corrosion Behavior, Mechanism, and Protection Strategies of Bronze Artefacts," *Heritage*, vol. 8, no. 8, p. 340, 2025.

²⁰ K. Sachdeva, M. Suri and S. Bhagat, "Comparison of Cleaning Treatments for Conservation and Restoration of Cotton, Wool and Silk Fabrics," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 3, no. VII, pp. 331-338, 2015.

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methods of cleaning are common in conservation field^{21,22,23,24}. The cleaning methods suggested to use in conservation of an object is selected according to the condition of the object²⁵.

4.1.1. Mechanical Cleaning

Due to the extremely fragile condition of the metallic threads, preliminary tests were conducted to evaluate their tolerance for mechanical cleaning. The initial stage involved removing loose surface dust and dirt using soft brushes to gently dislodge any particles adhering to the metallic threads. Additional tests assessed the feasibility of removing corrosion through gentle abrasion with fiberglass tools, in an attempt to clean the corrosion layers present on the metallic threads.

However, these preliminary trials proved unsuccessful and demonstrated that such methods posed a high risk of damage to the artifact, given its severe structural fragility, disintegration, and multiple tears. As a result, mechanical cleaning techniques were deemed inappropriate, and the decision was made to employ alternative cleaning approaches for the metallic threads, as outlined in the following sections.

4.1.2. Chemical Cleaning

Following the completion of mechanical cleaning, it became evident that this method was insufficient and unsuitable for removing corrosion products and embedded deposits on the metallic threads. Consequently, the decision was made to employ chemical cleaning techniques, as other advanced alternatives, such as laser cleaning, were not available²⁶. Chemical cleaning is one of the most widely applied methods for treating metals, with cleaning agents selected according to the artifact's condition, the type of corrosion present, and the intended treatment outcome.

²¹ S. Landi, *The Textile Conservator's Manual*, Butterworth & Heinemann., 1998.

²² A. Timar-Balazsy and D. Eastop, *Chemical Principles of Textile Conservation*, Oxford: Butter worth–Heinemann, 2002.

²³ B. Radojkovi', S. Risti', S. Polic', R. Jan'ci'c-Heinemann and D. Radovanovic', "Preliminary investigation on the use of the Q-switched Nd:YAG laser to clean corrosion products on museum embroidered textiles with metallic yarns," *Journal of culture Heritage*, vol. 23, pp. 128-137, 2017.

²⁴ D. Singhal and S. Bhagat, "Cleaning of textile artifacts in museums," *International Journal of Applied Home Science*, vol. 2, no. (7&8), pp. pp.237-242, 2015.

²⁵ O. Abdel Kareem and R. Alfaisal, "Treatment, Conservation and Restoration of the Bedouin died Textiles in the museum of Jordanian Heritage," *Mediterranean Archaeology and Archaeometry*, vol. 10, no. 1, p. 25-36, 2010.

²⁶ O. Abdel-Kareem and M. A. Harith, "Evaluating the Use of Laser Radiation in Cleaning of Copper Embroidery Threads on Archaeological Egyptian Textiles," *Applied Surface Science*, vol. 254, p. 5854–5860, 2008.

In this study, a specialized chemical cleaning solution was prepared using a mixture of Rochelle salt solution (5 g sodium hydroxide and 15 g sodium potassium tartrate), which acts as a mild alkaline complex helping to soften corrosion layers and loosen dirt deposits; sodium sesquicarbonate solution (2 g sodium carbonate and 2 g sodium bicarbonate), serving as a gentle alkaline buffer that stabilizes the pH and prevents sudden fluctuations; EDTA at 1%, functioning as a chelating agent that enhances the removal of corrosion products and inhibits further metal ion activity; Nonylphenol Ethoxylate (1%), providing mild surfactant properties that facilitate the detachment of surface contaminants; and isopropyl alcohol (10%), which acts as an organic solvent aiding in dissolving grease residues and ensuring rapid drying, thereby minimizing moisture-related risks. All components were diluted in distilled water to achieve an effective yet safe cleaning mixture for the metallic threads of the embroidered textile.

The solution was applied locally to the sirma threads using a white cotton swab wrapped around a wooden stick. The swab was immersed in the solution and gently rolled over the metallic sirma threads, with care taken to clean along the direction of the threads. Upon completion, residues were removed with a dry cotton swab, followed by rinsing with water and alcohol²⁷. The chemical cleaning procedure is illustrated in (Fig. 9).



**Fig. (9) Chemical cleaning of the sirma threads
and the textile of the table cover**

Subsequent examination of a metallic thread sample using Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX) revealed a marked improvement in surface condition, with the gold (Au) component becoming visible—previously obscured by a thick corrosion layer. The analysis confirmed the complete removal of corrosion deposits, resulting in a cleaner, more stable surface and increased relative concentrations of gold, silver, and copper compared to initial findings (Fig. 10).

²⁷ B. Miao, Z. Zhao, P. Guo, H. Li and Y. Wang, "Cleaning iron rust compounds from cotton textiles: application to Qing Dynasty armor," *Heritage Science*, vol. 11, 2023.

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Following the treatment of the metallic threads, the textile fibers underwent mechanical cleaning to remove stains. The process was performed using cotton and soft brushes with extreme caution²⁸.

Once cleaning of both metal threads and the textile were completed, the metal threads were selectively protected and isolated against the environmental conditions to be ready for any further treatment. The isolation and protection were carried out using Paraloid B-72 at 4% concentration, combined with benzotriazole at 1%. The treatment was applied locally with a soft brush to stabilize and protect the metal surfaces.

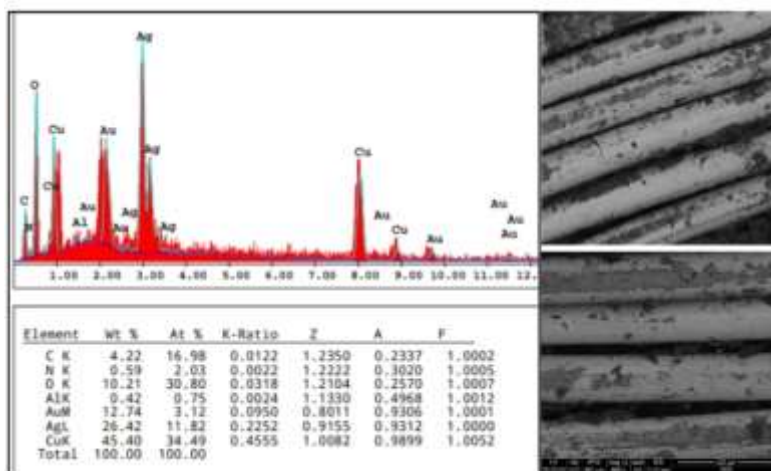


Fig. (10) SEM image with EDX analysis showing the removal of corrosion products from the sirma threads and the increased levels of gold, silver, and copper.

4.2 Consolidation process of the Artifact

Consolidation is one of the most critical stages of textile conservation, as most historic textiles suffer from significant fragility and structural weakness, necessitating careful reinforcement to ensure their long-term stability. In this study, the consolidation process was carried out in **two main phases**:

4.2.1 Reinforcement with Crepeline:

Since the outer borders of the tablecloth were in good condition and showed no material loss, a piece of silk crepeline measuring 35×110 cm was prepared to support the inner embroidered section only, which exhibited extensive tears and losses in the ground fabric. The crepeline was positioned on the reverse side of the artifact, and Klucel G adhesive was applied using a fine brush to secure localized areas.

Reinforcement was limited only to loose and partially detached sections that were at risk of further loss, avoiding full-surface adhesion. This minimal intervention approach was adopted to ensure that the textile remains easy to reverse or retreat in the future, should additional conservation be required. The treatment steps are illustrated in the accompanying figures.

4.2.2 Reinforcement with linen support

The most effective method for stabilizing weak textiles is to provide adequate support by lining them with a suitable fabric, which ensures additional reinforcement and

²⁸ O. Abdel-Kareem, Principles of Conservation and Restoration of Archaeological Organic Materials, textiles, wood, leather, manuscripts, boons, basketry and feathers, Cairo: Cairo University Center for the Printing and the Publishing, 2015, p. 428.

stabilization²⁹. Mounting the artifact on a new support fabric is a well-established practice that enhances structural stability and also facilitates safe exhibition and storage³⁰. recommends the use of inert natural linen as the most appropriate choice for this purpose, owing to its breathability, durability, and low susceptibility to biological deterioration.

In this study, after reinforcing the textile with crepeline, a custom support system was prepared. The mount consisted of a wooden frame made from acid-free pinewood, measuring 75 × 165 cm, with dimensions exceeding the artifact's size by approximately 10 cm on each side. The frame was sanded to prevent snagging of fibers during tensioning and treated with fungicidal and insecticidal agents to ensure long-term preservation. A piece of plain-weave raw linen (1:1), measuring 100 × 190 cm, was selected as the primary support fabric. The linen was first boiled in a neutral soap solution for 20 minutes to remove starches or impurities, then thoroughly rinsed, air-dried at room temperature, and ironed. Finally, it was stretched tightly over the wooden frame and secured with stainless steel nails, ensuring proper alignment of the warp threads vertically and the weft threads horizontally.

The artifact was carefully aligned and placed on the mounted linen, maintaining a 10 cm margin on all sides. The textile was first secured with a temporary basting stitch (running stitch), followed by permanent stitching using natural silk thread chosen to match the original color of the textile and the linen support. A combination of blanket stitch, overcast stitch, herringbone stitch, and couching stitch was employed to secure the edges of missing areas, stabilize the outer borders, reinforce the overall structure, and anchor the metallic embroidery threads. Upon completion of this step, the restoration process was finalized (Fig. 11).

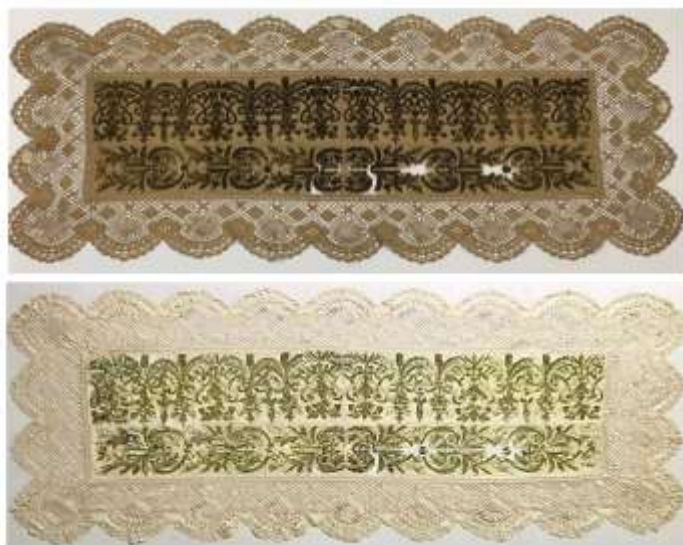


Fig. (11) The historic table cover before and after conservation and restoration treatments.

5. Proposed Display Plan for the table cover:

Following the conservation and restoration of historic textiles, one of the primary challenges is ensuring their long-term preservation, particularly because organic materials, such as textiles, are highly sensitive to fluctuations in humidity and

²⁹ O. Abdel-Kareem and R. Alfaisal, "Treatment, Conservation and Restoration of the Bedouin died Textiles in the museum of Jordanian Heritage," *Mediterranean Archaeology and Archaeometry*, vol. 10, no. 1, p. 25-36, 2010.

³⁰ S. Landi, *The Textile Conservator's Manual*, Butterworth & Heinemann., 1998.

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temperature and are vulnerable to fungal growth, bacterial contamination, and insect damage.

To guarantee the longevity of the artifact, it is essential to provide a controlled and secure display environment within museum galleries or storage facilities. Proper display methods must safeguard the textile while maintaining optimal environmental conditions, including stable temperature and relative humidity levels and the use of appropriate, non-damaging lighting systems.

The textile can be displayed using two approaches:

1. Horizontal display in a showcase equipped with temperature and humidity monitoring devices, with regular inspection and documentation of readings.
2. Vertical display in a showcase fitted with environmental control systems to regulate temperature and humidity, along with suitable lighting arrangements to minimize photodegradation risks.

6. Conclusion

This study demonstrates that the conservation of the historic crochet table cover embroidered with metallic threads required a carefully designed approach that integrated comprehensive scientific analysis with specialized conservation techniques to achieve stable and sustainable results. Advanced analytical methods, including Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDX), enabled the precise identification of natural fibers, metallic threads, and corrosion compounds, providing critical insights into the material composition and causes of deterioration. These findings informed the careful selection of treatment materials and methods to ensure optimal results while maintaining the artifact's authenticity.

The conservation plan successfully restored the structural stability of the textile, reinforced its weakened sections, and revealed decorative details previously concealed beneath layers of dirt and corrosion. All treatments adhered to the principles of minimal intervention and employed inert, reversible materials, allowing for future re-treatment without compromising the artifact.

This research emphasizes the necessity of implementing preventive conservation measures, including environmental control (temperature and relative humidity) and appropriate display and storage strategies, to protect historic textiles from further deterioration. Moreover, the study presents a practical conservation model for composite textiles combining organic and metallic elements and highlights the importance of interdisciplinary collaboration among specialists in analytical science, textile conservation, and preventive care to ensure the long-term preservation of textile heritage as a key element of cultural and historical identity.

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